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Privacy and the Medical Device Regulation (EU) 2017/745

Universiteit Leiden
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LUMC Leiden University Medical Center

FUTURE OF PRIVACY FORUM

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Privacy and the Medical Device Regulation

- MDR, GDPR, and a Proposal for AI Regulation
- Principles relating to processing of personal data
- Privacy framing for AI issues
- Practical do's and don'ts

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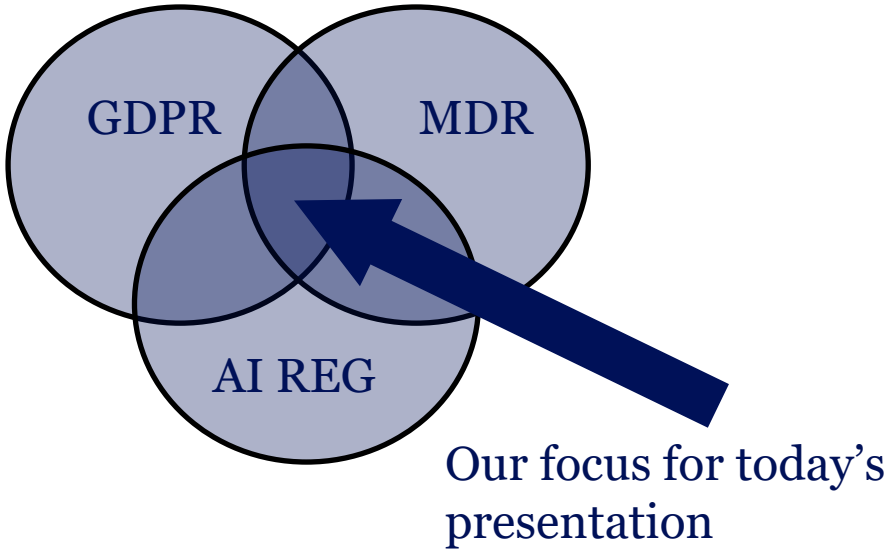
MDR, GDPR, and a Proposal for AI Regulation

- The **MDR** contains very few privacy nor data protection articles
  - Article 110 refers to the GDPR
  - Article 1(55) defines 'informed consent' (i.e., corresponds to GDPR consent to some extent)
  - Article 63 details informed consent
- The **GDPR** is a horizontal legislation
- The **Proposal for AI Regulation** is specific to AI (systems)



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MDR [1]

- MDR subject matter and scope:
  - lays down rules concerning the **placing on the market**, **making available on the market** or **putting into service** of medical devices for human use and accessories for such devices in the Union.
  - This Regulation also applies to clinical investigations concerning such medical devices and accessories conducted in the Union.
  - Art 1(4): For the purposes of this Regulation, medical devices, accessories for **medical devices, and products listed in Annex XVI** to which this Regulation applies pursuant to paragraph 2 shall hereinafter be referred to as '**devices**'.

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MDR [2]

- Art 1(40): '**conformity assessment**' means the process demonstrating whether the requirements of this Regulation relating to a device have been fulfilled.
- Art 1(43) '**CE marking of conformity**' or '**CE marking**' means a marking by which a manufacturer indicates that a device is in conformity with the applicable requirements set out in this Regulation and other applicable Union harmonisation legislation providing for its affixing.
- Annex I, Chapter 1, General Requirements (2)
  - The requirement in this Annex **to reduce risks as far as possible** means **the reduction of risks as far as possible without adversely affecting the benefit-risk ratio**.

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Proposal for AI Regulation

- Proposal for a Regulation laying down harmonised rules on artificial intelligence
  - **Similar to MDR, it includes conformity assessments**
  - Rec 64 '(…) the conformity assessment of such **[i.e., high risk AI] systems** should be carried out as a general rule by the provider under its own responsibility (…)'
- Additional obligations include:
  - Necessary corrective actions – ex post if non conformity [Art 16 (g) + 21]
  - Duty of information after placed on the market to: national authorities and notified body if non compliance / corrective actions taken [Art 16 (h) + 22]
  - Demonstration of conformity upon request & cooperation with any national competent authorities [Art 16 (j) + 23]

URL: https://eur-lex.europa.eu/legal-content/EN/TXT/?qid=1623335154975&uri=CELEX%3A52021PC0206

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General Data Protection Regulation

- Article 5 - Principles relating to processing of personal data
- Article 5(1)(a) - GDPR lawfulness, fairness and transparency
- Article 5(1)(b) - Purpose limitation
- Article 5(1)(c) - Data minimisation
- Article 5(1)(d) - Accuracy
- Article 5(1)(e) - Storage limitation
- Article 5(1)(f) - Integrity and confidentiality
- Article 5(2) - Accountability

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Privacy: by design and by default (Article 25 GDPR)

- (…) implement appropriate technical and organisational measures, such as pseudonymisation, which are designed **to implement data-protection principles, such as data minimisation**, in an effective manner (…)
- (…) implement appropriate technical and organisational measures for ensuring that, **by default**, only personal data which are **necessary for each specific purpose** of the processing are processed. That obligation applies to the amount of personal data collected, the extent of their processing, the period of their storage and their accessibility (…)

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Privacy: a risk based approach (Recital 75 GDPR)

- The risk to the **rights and freedoms of natural persons**,
- of varying likelihood and severity, may result from personal data processing which could lead to
  - physical,
  - material or
  - non-material damage.

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Privacy: a risk based approach (Recital 76 GDPR)

- The **likelihood** and **severity** of the risk to the rights and freedoms of the data subject **should be determined** by reference to the nature, scope, context and purposes of the processing.

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Privacy: a risk based approach (Recital 83 GDPR)

- In order to maintain security and to prevent processing in infringement of this Regulation, the controller or processor should **evaluate the risks** inherent in the processing and **implement measures to mitigate those risks**, such as encryption.

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Privacy framing for AI issues – “FAT” (1)

- The discussion of AI privacy implications has largely been framed around the concepts of **Fairness**, **Accountability**, and **Transparency** (frequently shortened to “FAT”).
- Taken together, these concepts represent some of the unique aspects of machine learning models that raise the most concern.

[Future of Privacy Forum (2018). *The Expert’s Guide To Artificial Intelligence and Machine Learning,* p. 22]

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Privacy framing for AI issues – “FAT” (2)

- **Fairness**: Ensure that algorithmic decisions do not create discriminatory or unjust impacts when comparing across different demographics or affected communities and individuals.
- **Accountability**: Algorithmically informed decisions have the potential for significant social impact, and must be designed and implemented in publicly accountable ways, such as an obligation to report, explain and justify specific decisions as well as mitigate negative impacts and potential harms.
- **Transparency**: ensuring that the personal data collected, the model’s function and process, and the role of the model’s output in final decision-making are explainable to the user or individual impacted by the decision.

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Privacy framing for AI issues – “FAT” (3)

- Fairness is a highly prized human value.
- **Inequality** and **unfairness** have complex causes, but bias in the decisions that organizations make about individuals is often a key aspect.
- **Transparency** is key in building and maintaining trust.
- Requirements for transparency:
  - Analytical techniques to expose **bias**,
  - **Ethical** and **legal considersations** that inform best responses.

[Centre for data ethics and innovation (2020). *Review in to bias in algorithmic decision-making*, pp. 3-13, 21-36]

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Explainability and transparency (1)

- While not interchangeable, these two concepts are related.
- **Explainability** describes the privacy risk discussions that need to occur when decisions are made in the development and design stages of a ML model.
- When decisions made by the model will affect individuals or members of a group, and are reached based solely on the model output, without human review or intervention, there is a potential for problems.

[Future of Privacy Forum (2018). *The Expert's Guide To Artificial Intelligence and Machine Learning,* p. 17]

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Explainability and transparency (2)

- **Transparency** is generally the ability to interpret the functioning of machine learning models, provide explanations for specific algorithmic outputs, and describe or provide frameworks for conducting ethical and legal reviews and audits.
- It might include providing data subjects with process details, informing them when a decision was reached via an automated system, and providing an understanding of the safeguards and rights available to them.

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Explainability and transparency (3)

- The reality is that many cutting edge machine learning models include multi-layer neural networks which follow as an internal process under which a particular outcome may not be able to be understood in a mathematically detailed, replicable way even by the data scientists and designers.
- But the multi-layer feature in fact may make the model “better” in terms or relevance or applicability.
- This is recognized as the accuracy vs. interpretability tradeoff (in terms of explainability and transparency).
- It is also sometimes called **the “black box” problem.**

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FAT/ML

- Principles for Accountable Algorithms and a Social Impact Statement for Algorithms

URL: https://www.fatml.org/resources/principles-for-accountable-algorithms.

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FAT/ML

- Principles for Accountable Algorithms and a Social Impact Statement for Algorithms

Explainability

Ensure that algorithmic decisions as well as any data driving those decisions can be explained to end-users and other stakeholders in non-technical terms.

URL: https://www.fatml.org/resources/principles-for-accountable-algorithms.

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FAT/ML

- Principle
  Impact S

Explainability

Ensure that algorithmic decisions as
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Explainability

Guiding Questions

- Who are your end-users and stakeholders?
- How much of your system / algorithm can you explain to your users and stakeholders?
- How much of the data sources can you disclose?

Initial Steps to Take

- Have a plan for how decisions will be explained to users and subjects of those decisions. In some cases it may be appropriate to develop an automated explanation for each decision.
- Allow data subjects visibility into the data you store about them and access to a process in order to change it.
- If you are using a machine-learning model:
  - consider whether a directly interpretable or explainable model can be used.
  - describe the training data including how, when, and why it was collected and sampled.
  - describe how and when test data about an individual that is used to make a decision is collected or inferred.
- Disclose the sources of any data used and as much as possible about the specific attributes of the data. Explain how the data was cleaned or otherwise transformed.

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Explainability

Fairness

Ensure that algorithmic decisions do not create discriminatory or unjust impacts when comparing across different demographics (e.g. race, sex, etc).

We have left some of the terms above purposefully under-specified to allow these principles to be broadly applicable. Applying these principles well should include understanding them within a specific context. We also suggest that these issues be revisited and discussed throughout the design, implementation, and release phases of development. Two important principles for consideration were purposefully left off of this list as they are well-covered elsewhere: privacy and the impact of human experimentation. We encourage you to incorporate those issues into your overall assessment of algorithmic accountability as well.

URL: https://www.fatml.org/resources/principles-for-accountable-algorithms.

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FAT/ML

- Princip
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Fairness

Guiding Questions

- Are there particular groups which may be advantaged or disadvantaged, in the context in which you are deploying, by the algorithm / system you are building?
- What is the potential damaging effect of uncertainty / errors to different groups?

Initial Steps to Take

- Talk to people who are familiar with the subtle social context in which you are deploying. For example, you should consider whether the following aspects of people's identities will have impacts on their equitable access to and results from your system:
- Race
- Sex
- Gender identity
- Ability status
- Socio-economic status
- Education level
- Religion
- Country of origin
- If you are building an automated decision-making tool, you should deploy a fairness-aware data mining algorithm. (See, e.g., the resources gathered at http://fatml.org).
- Calculate the error rates and types (e.g., false positives vs. false negatives) for different sub-populations and assess the potential differential impacts.

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Tools for Explainability [1]

Triggers	User/Learner's Goal	
	Need to Understand	Need to Accomplish
How do I use it?	Use	Primary task goals
How does it work?	Mechanism, Understandability	Feeling of Satisfaction
What did it just do?	Mechanism, Understandability	Feeling of Sufficiency
What does it achieve?	Function	Usefulness
What will it do next?	Observability	Trust, Reliance
How much effort will this take?	Usability	Primary Task Goals
What do I do if it gets it wrong?	Anticipation	Control
How do I avoid the failure modes?	Surprise	Adaptation, Work-around
What would it have done if x was different?	Mechanism, Understandability	Curiosity, Trust, Reliance
Why didn't it do z?	Anticipation, Surprise	Curiosity, Trust, Reliance

Mueller, S. T., Hoffman, R. R., Clancey, W., Emrey, A., & Klein, G. (2019). Explanation in human-AI systems: A literature meta-review, synopsis of key ideas and publications, and bibliography for explainable A, p. 90.

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Tools for Explainability [2]

Format (How it is expressed)	Reference (What it is about)
• Visualizations (e.g., Heat maps) • Text (Statements, Narratives or Stories, Answers to queries, Human-machine dialogs) • Formal Expressions (Logical expressions, Matrices) • Conceptual Process Models (Diagrams) • Graphs, Networks • Tables • Abstractions, generalizations • Timelines • Hierarchies (Trees)	• Examples (includes misclassifications, counter-examples, outliers, clear cases, close competitors) • Patterns, Classes, Ontologies • Features, Weights, Probabilities, Ranks, Parameters • Decisions, Strategies, Goals • Algorithms, Computational Processes, Proofs • Incidents, Events (includes self-explanations or stories) • Cause-effect relations

Mueller, S. T., Hoffman, R. R., Clancey, W., Emrey, A., & Klein, G. (2019). Explanation in human-AI systems: A literature meta-review, synopsis of key ideas and publications, and bibliography for explainable A, p. 84.

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Autoriteit Consument en Markt (10 Dec 2020)

Onderzoek naar toezicht op algoritmische toepassingen

Dit position paper is een vertrekpunt van waaruit de ACM het toezicht op algoritmische toepassingen verder wil ontwikkelen. Het biedt algemene handvatten voor onderzoeken naar overtredingen waarbij algoritmische toepassingen een rol spelen.

De ACM beschrijft in het position paper wat algoritmische toepassing zijn en wanneerzij relevant zijn voor de ACM. Vervolgens licht de ACM toe hoe zij in de praktijk algoritmische toepassingen kan onderzoeken en bespreekt zij enkele uitdagingen waar zij in de praktijk mee te maken kan krijgen.

De ACM draagt met deze publicatie bij aan de nationale en internationale discussie over de toepassing van AI (artificiële intelligentie) door bedrijven en het toezicht daarop door toezichthouders.

Wachter S., Mittelstat B. & Rusell C. (2021). Bias Preservation in Machine Learning: The Legality of Fairness Metrics Under EU Non-Discrimination Law. In: *West Virginia Law Review*, *Forthcoming*.

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Autoriteit Co 20)

Fairness metrics	Bias preserving?
1. Group fairness, Statistical (demographic) parity[i]	✗
2. Conditional statistical (demographic) parity, Conditional independence[ii]	✗
3. Predictive parity, outcome test[iii]	✓
4. False positive error rate balance[iv]	✓
5. False negative error rate balance,[v] Equal opportunity[vi]	✓
6. Equalized odds[vii]	✓
7. Conditional use accuracy equality[viii]	✓
8. Overall accuracy equality[ix]	✓
9. Treatment equality[x]	✓
10. Test-fairness or calibration[xi]	✓
11. Well-calibration[xii]	✓
12. Balance for positive class[xiii]	✓
13. Balance for negative class[xiv]	✓
14. Causal discrimination[xv] (direct discrimination)	*
15. Fairness through unawareness[xvi]	*
16. Fairness through awareness[xvii]	✗
17. Counterfactual fairness[xviii]	✗
18. No unresolved discrimination[xix] (causal variant of 2)	✗
19. No proxy discrimination[xx]	✗
20. Path based causal reasoning[xxi]	✗

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Ten Practical do's and don'ts

- Privacy is a fundamental right (see also Article 12 – 22 GDPR)
- Select before you collect (data minimization)
- Use anonymization techniques where possible
- Use of pseudonymization where possible
- Design for privacy preserving data sharing
- Use encryption where possible
- Design for granular consent controls including the revocation of consent
- Minimize data retention and minimize the persistency of identifiers
- Do not forget data subject's rights (DSARS)
- Design for data portability (if applicable)

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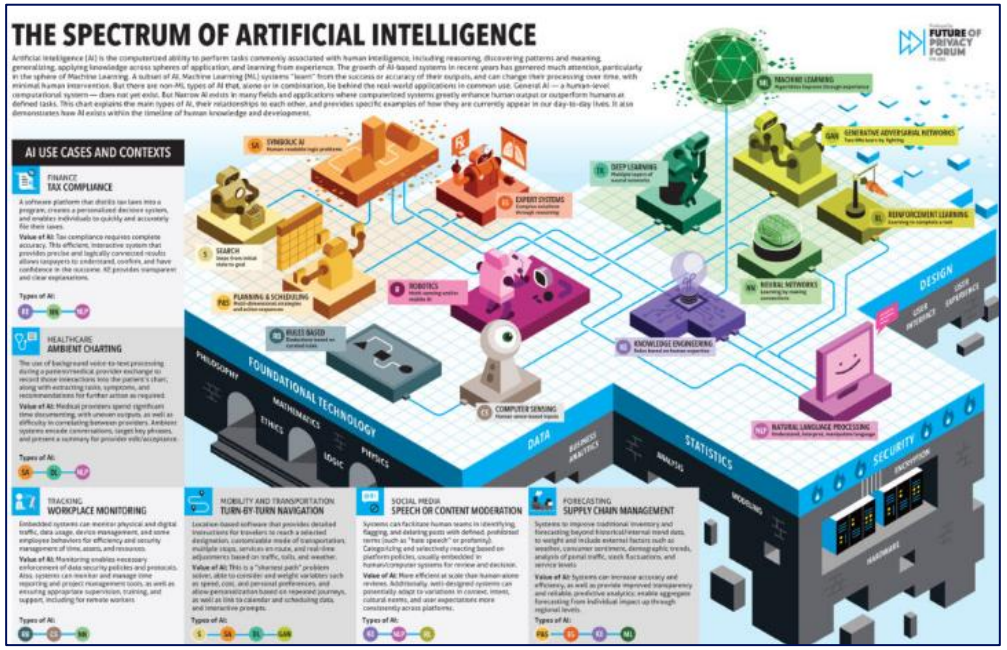
Three Resources

- **Infographic: The Spectrum of Artificial Intelligence**
  https://fpf.org/wpcontent/uploads/2021/01/FPF_AIEcosystem_illo_03.pdf
- **Machine Learning and Speech: Slides and Recording of FPF's Digital Data Flows Masterclass.**
  https://fpf.org/blog/machine-learning-and-speech-a-review-of-fpfs-digital-data-flows-masterclass/
- **The Privacy Expert's Guide to AI and Machine Learning**
  https://fpf.org/wp-content/uploads/2018/10/FPF_Artificial-Intelligence_Digital.pdf



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Thank you for your attention

Contact: rvaneijk@fpf.org

To learn more about FPF's Understanding Digital Data Flows, please visit **fpf.org/training**.

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